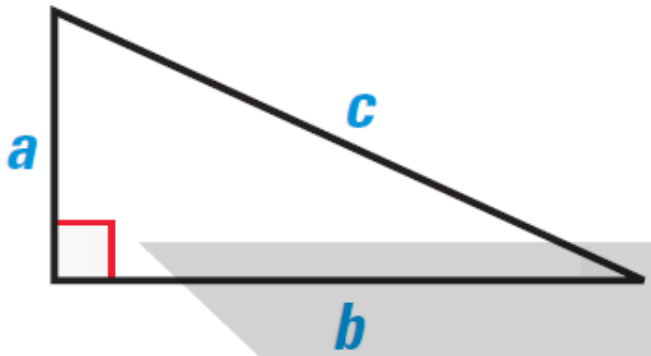
1. Trigonometry

1.1. Pythagoras Theorem

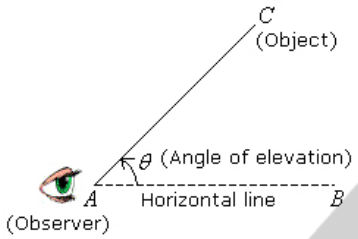
- To find the Hypotenuse
  - $c^2 = a^2 + b^2$



- To find one of the shorter sides
  - $a^2 = c^2 - b^2$
  - $b^2 = c^2 - a^2$

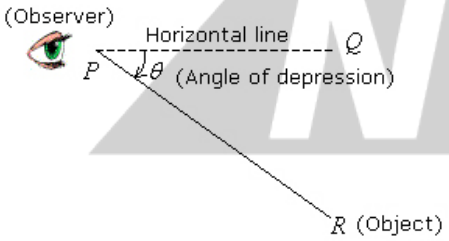
Angle of Elevation

- Angle above the horizontal line



Angle of Depression

- Angle below the horizontal line.



Area of a Right-Angled Triangle

$$\text{Area} = \frac{1}{2}bh$$

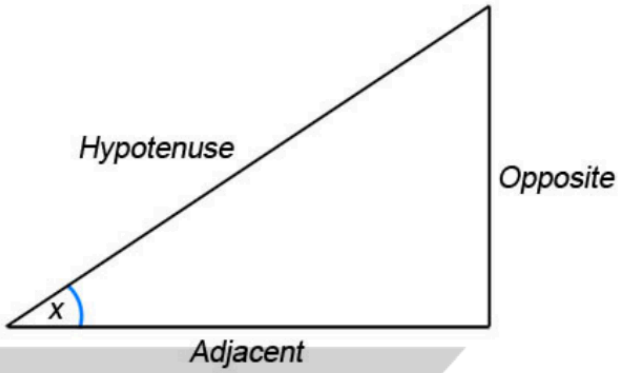
- b = base
- h = height (a in the triangle above)

1.2. Ratios

- The perpendicular distance from a point to a line is the shortest

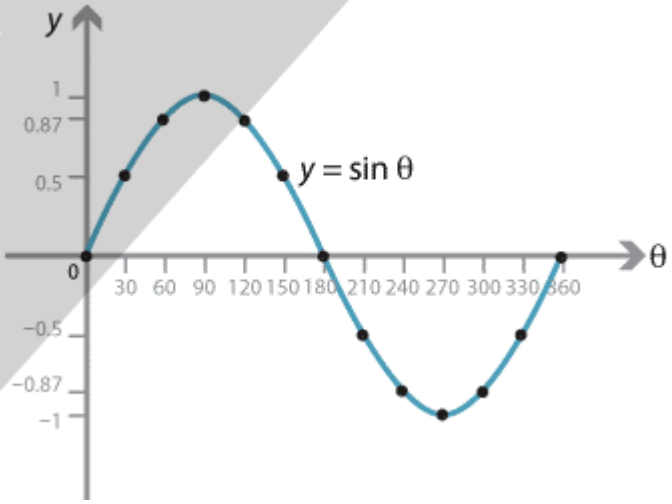
Right-Angled Triangles

- $\sin x = \frac{\text{Opposite}}{\text{Hypotenuse}}$ → SOH
- $\cos x = \frac{\text{Adjacent}}{\text{Hypotenuse}}$ → CAH
- $\tan x = \frac{\text{Opposite}}{\text{Adjacent}}$ → TOA

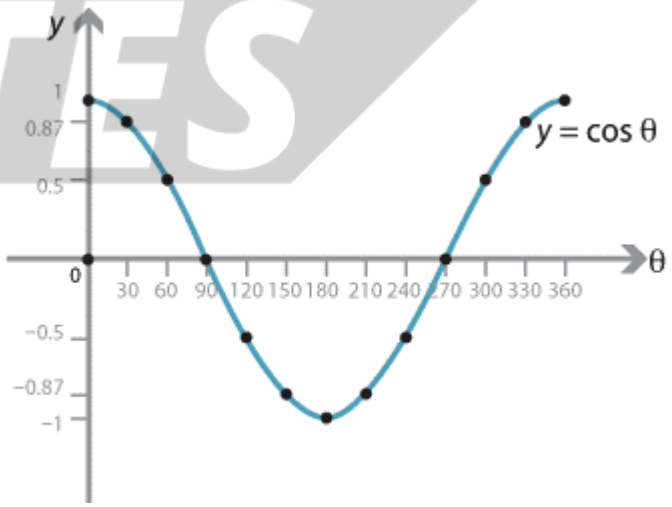


1.3. Graphs of Simple Trigonometric Functions

- $\sin(x) = \sin(180^\circ - x)$



- $\cos(x) = \cos(360^\circ - x)$



- Sine and cosine shifted by 90°
- Sine has x-intercepts at multiples 180°, and cosine at (90° + multiples of 180°)

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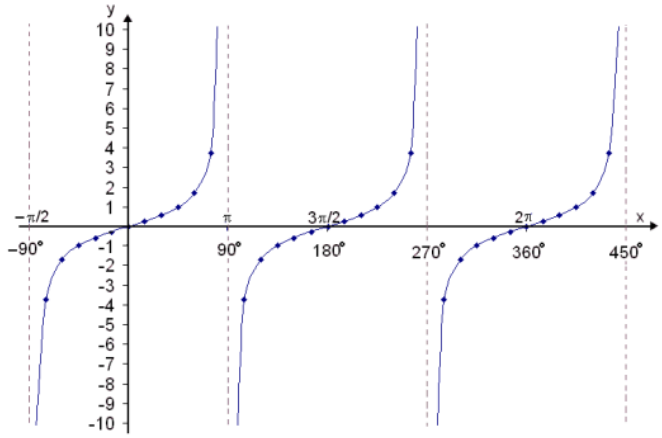
- $\tan(x) = \tan(180^\circ + x)$



- Goes to infinity at 90°, 270°, 450°, ...
- The tan graph has asymptotes at 90, 270, 450
- Has x-intercepts at multiples of 180°

	0	30	45	60	90
Sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
Tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	Doesn't Exist

1.4. Sine & Cosine Rules

Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine Rule

- To find the angle given 3 sides:

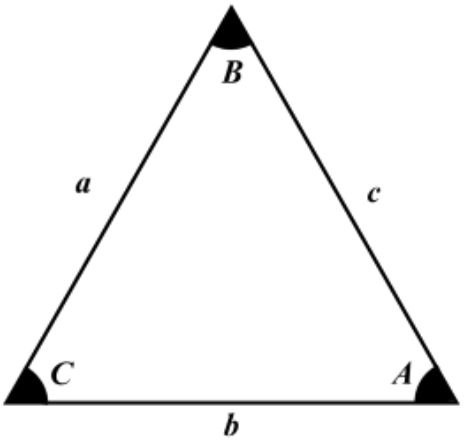
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- To find the side given angle and two sides

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Area of a Non-right-angled Triangle

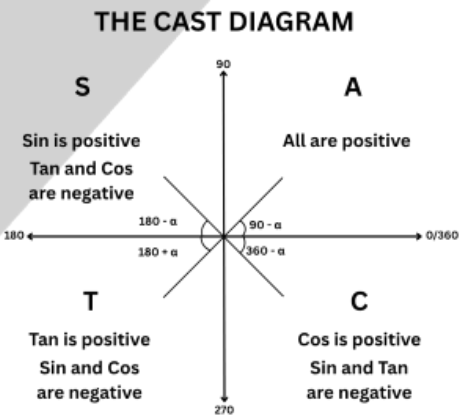
$$\frac{1}{2}ab\sin C$$



1.5. Solving Trigonometric Examples

Trigonometric Equations

- CAST diagram helps us to see in which quadrants the trig ratios are positive



Q.2 $\cos(x) + 1 = 0$ for $0 \le x \le 360$

Step 1: Isolate $\cos(x)$

$2\cos(x) + 1 = 0$

$2\cos(x) = -1$

$\cos(x) = \frac{-1}{2}$

Step 2: Find base angle

$\cos(\alpha) = \frac{1}{2}$ (Ignore negative sign for base angle)

$\alpha = \cos^{-1}(\frac{1}{2})$

$\alpha = 60$

Step 3: Determine the quadrants where the angle can lie

Use CAST diagram to identify quadrants

Since $\cos(x)$ was negative, we look for quadrants where cosine is negative

The solutions must lie in Quadrants II and III.

Step 4: Form angles in those quadrants using the base angles

Quadrant II

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$x = 180^\circ - \alpha$

$x = 180^\circ - 60^\circ = 120^\circ$

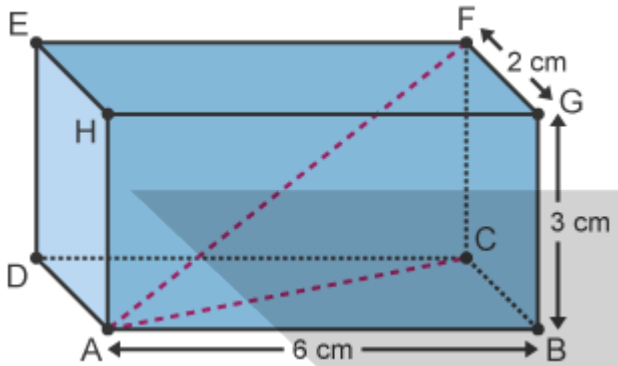
Quadrant III

$x = 180^\circ + \alpha$

$x = 180^\circ + 60^\circ = 240^\circ$

Final Answer: $\boxed{x = 120^\circ, 240^\circ}$

Pythagoras Theorem in 3D Shapes



- $2^2 + 6^2 = AC^2$
- AC = 6.32
- $6.32^2 + 3^2 = AF^2$
- AF = 7 (1sf)

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